



9-Month Diploma

Bidz: A Secure Web-Based Auction Platform with Real-Time Bidding and Transaction Control

Track: Software Development

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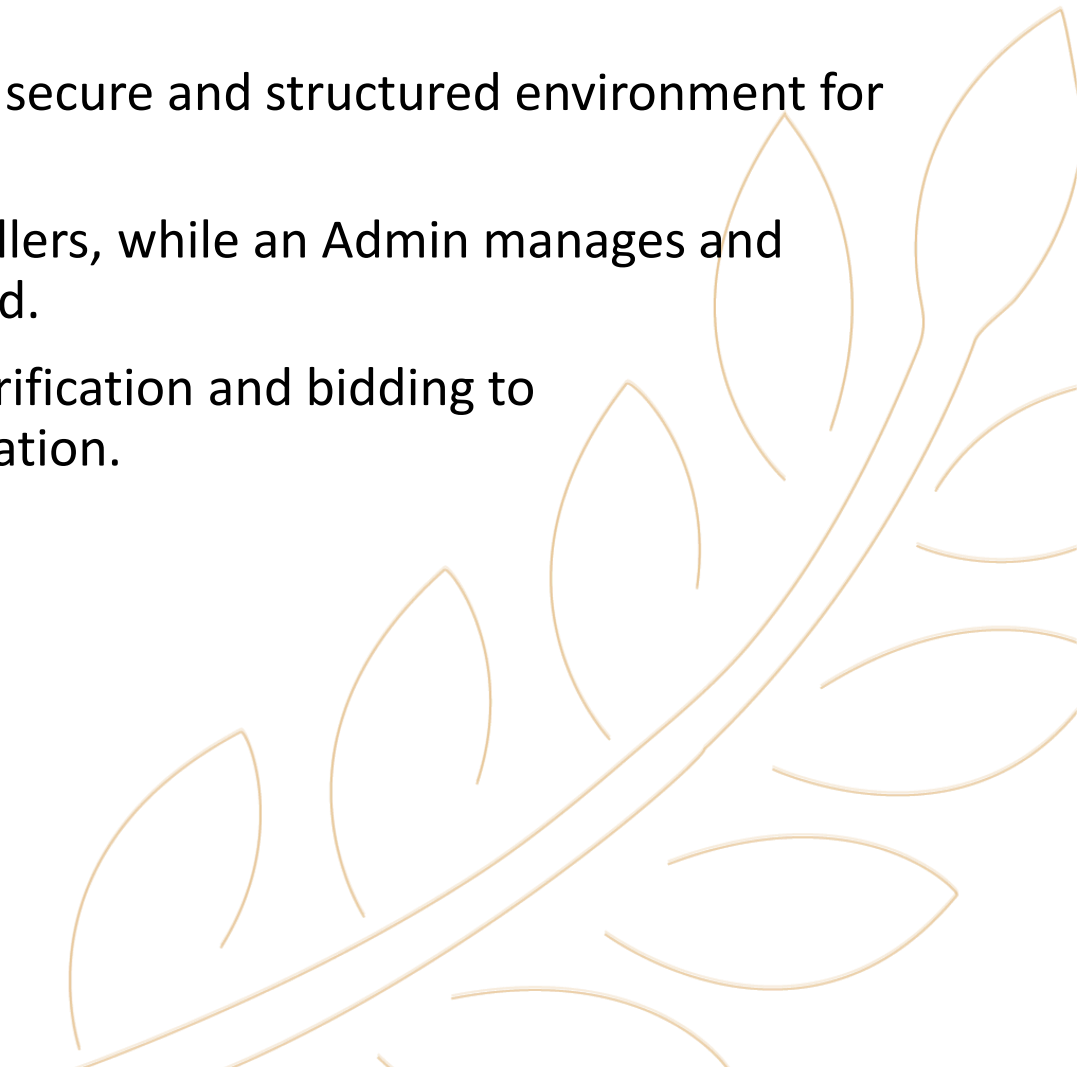
Agenda



- Introduction
- Problem Definition
- Motivation & Objectives
- Implementation (Seller & Customer Flows)
- System Architecture
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- Future Work



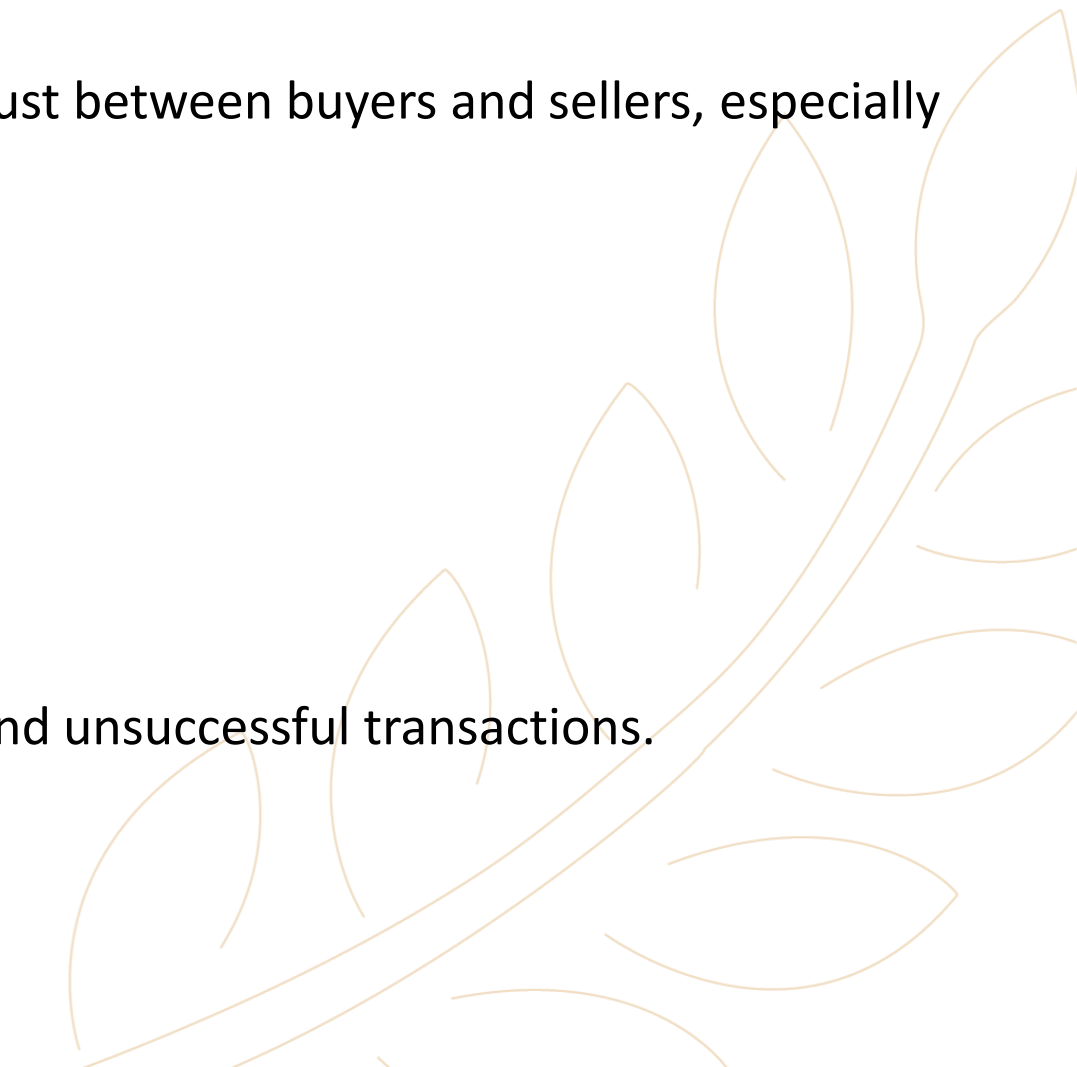
Introduction



- **Bidz** is a web-based auction platform designed to provide a secure and structured environment for buying and selling products through online auctions.
- The platform allows users to participate as Customers or Sellers, while an Admin manages and monitors the entire platform through a dedicated dashboard.
- **Bidz** covers the complete auction lifecycle, from product verification and bidding to communication, transaction handling, and delivery confirmation.

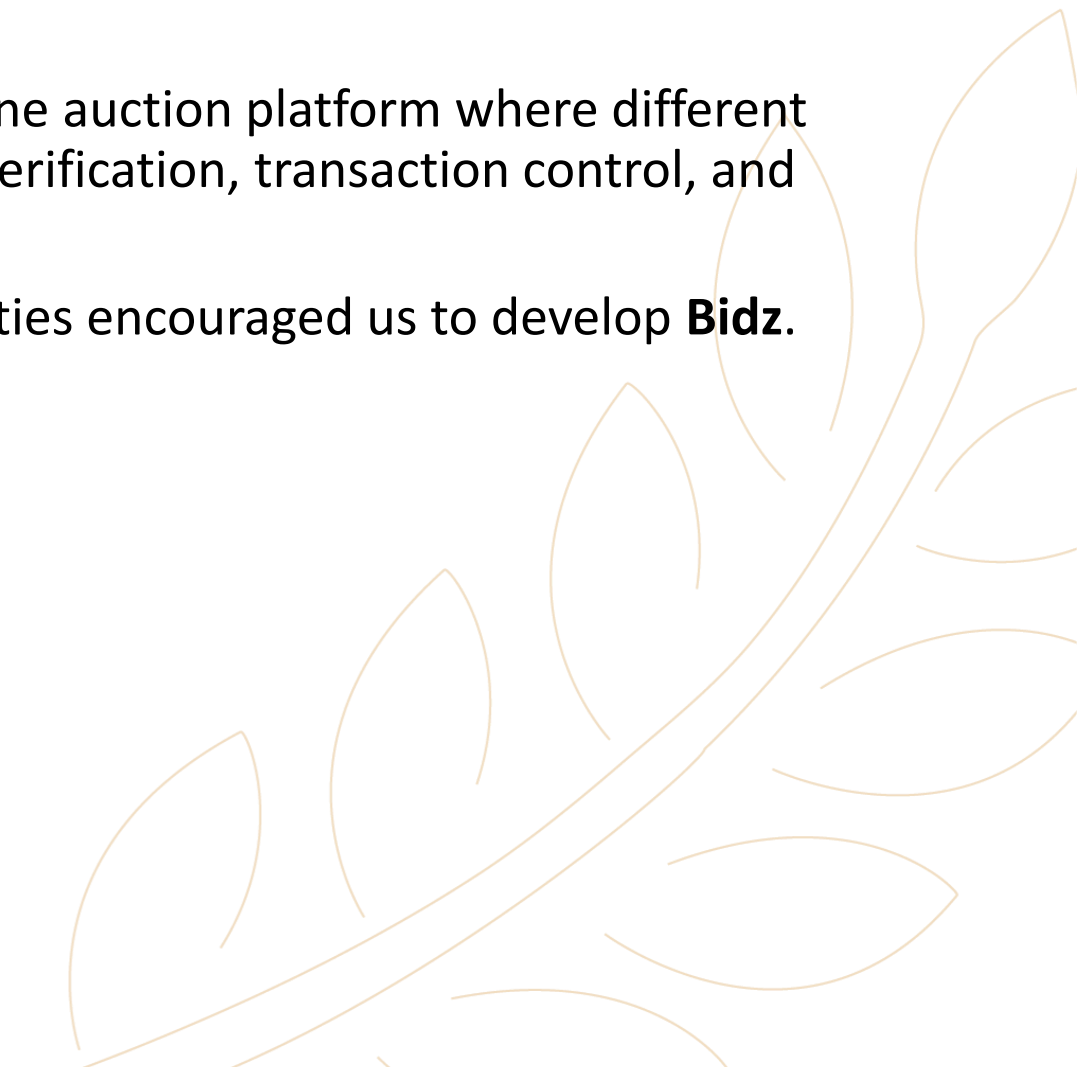
Problem Definition



- Online auction platforms can face challenges that reduce trust between buyers and sellers, especially when dealing with valuable or diverse products.
 - The main problems addressed by **Bidz** include:
 - Fake or inaccurate product information.
 - Insufficient product and document verification
 - Unsafe delivery confirmation
 - Lack of secure payment holding
 - Limited communication between buyers and sellers
 - Lack of an effective reporting and moderation mechanism
 - These challenges may increase the risk of fraud, disputes, and unsuccessful transactions.
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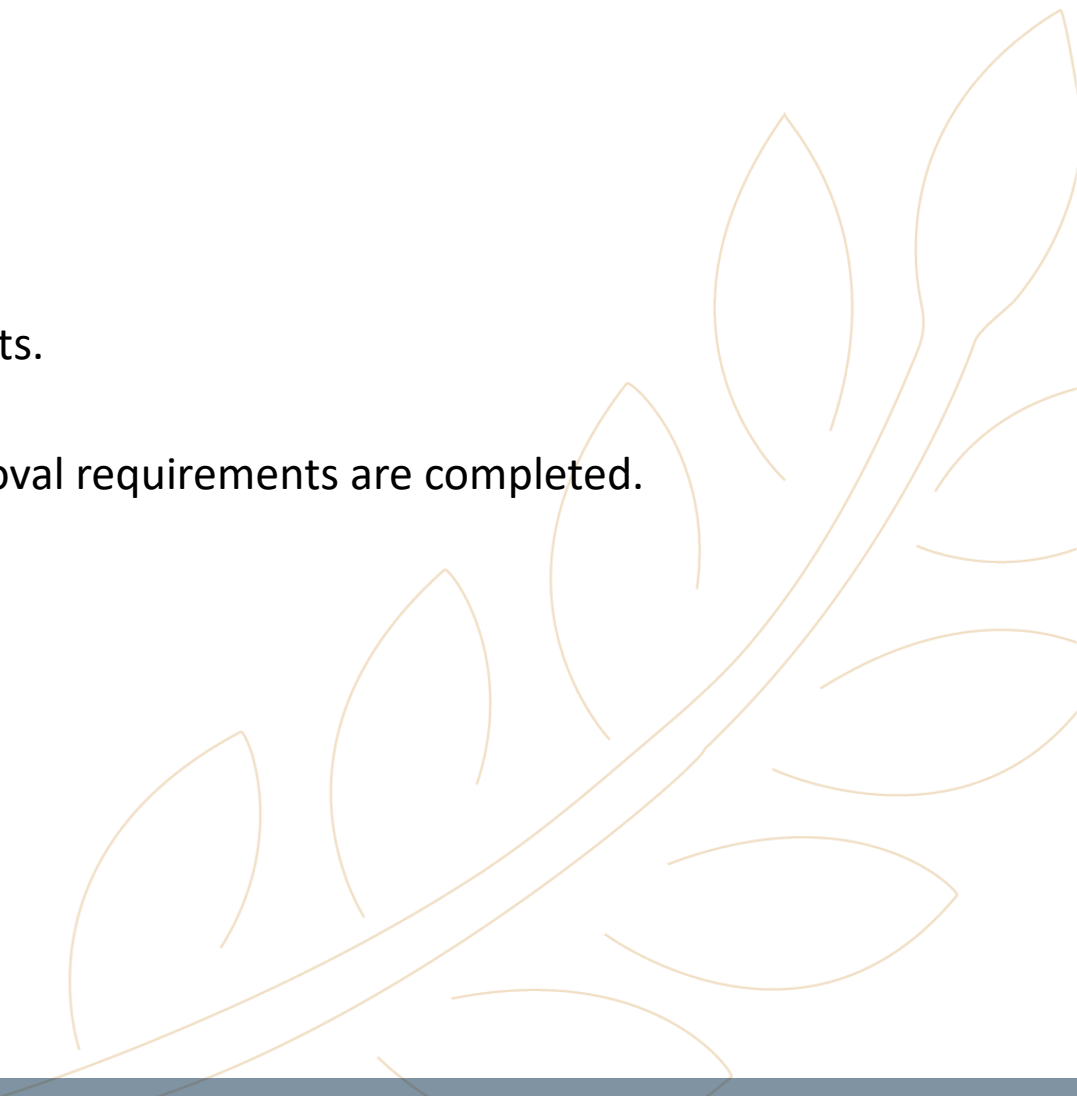
Motivation & Objectives

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- **Bidz** was motivated by the need for a secure and flexible online auction platform where different types of products can be offered while maintaining product verification, transaction control, and communication between users.
 - The limited availability of platforms combining these capabilities encouraged us to develop **Bidz**.
 - **Bidz** aims to:
 - Provide a secure online auction environment.
 - Allow users to act as both Customers and Sellers.
 - Verify product information and legal documentation.
 - Support transparent and competitive bidding.
 - Enable real-time communication between bidders and sellers.
 - Provide reporting and administrative moderation.
 - Protect users and APIs through multiple security mechanisms.
 - Control transaction funds until delivery confirmation.
 - Automate auction expiration and management.
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Implementation (1/2)



- **Bidz** implements a complete auction workflow.
 - Seller Side:
 - Purchase an auction ticket with a duration from 1 to 31 days.
 - Submit the product details, images, and required legal documents.
 - Obtain product approval.
 - The auction becomes active only after both the Ticket and Approval requirements are completed.
 - Receive bids from customers.
 - Communicate individually with bidders using real-time chat.
 - Select a bid and close the deal.
 - Complete the delivery confirmation process.
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Implementation (2/2)



- Customer Side:
- Browse available auctions.
 - Select a product.
 - Place a bid higher than the current bid.
 - Receive updates when another bidder places a higher offer.
 - Communicate with the seller.
 - Complete the transaction if the seller selects the bid.



Architecture (1/2)

- **Bidz** follows a Web Application architecture consisting of a React frontend, Node.js/Express backend, and MongoDB database.
- Frontend Main Components:
 - React
 - User and Admin interfaces
- Backend Main Components:
 - Node.js
 - Express.js
 - REST APIs
 - Business logic
 - Authentication & Authorization
 - Auction and transaction management

Architecture (2/2)

- Database Main Components:

- MongoDB
- Mongoose

- Real-Time Communication:

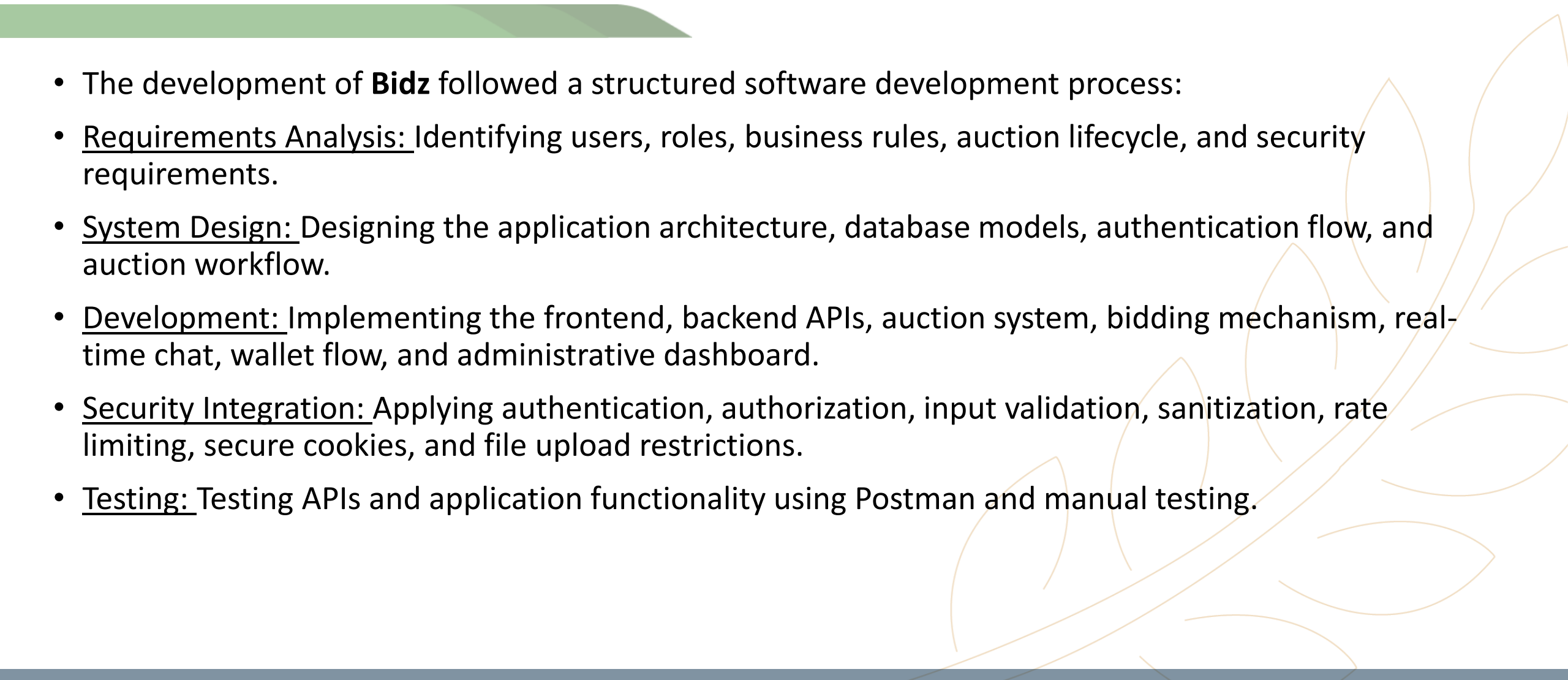
- Socket.IO

- Security Layer:

- JWT
- HttpOnly Cookies
- RBAC
- Middleware
- Input Validation
- Sanitization
- Rate Limiting
- Helmet / CSP / CORS
- File validation

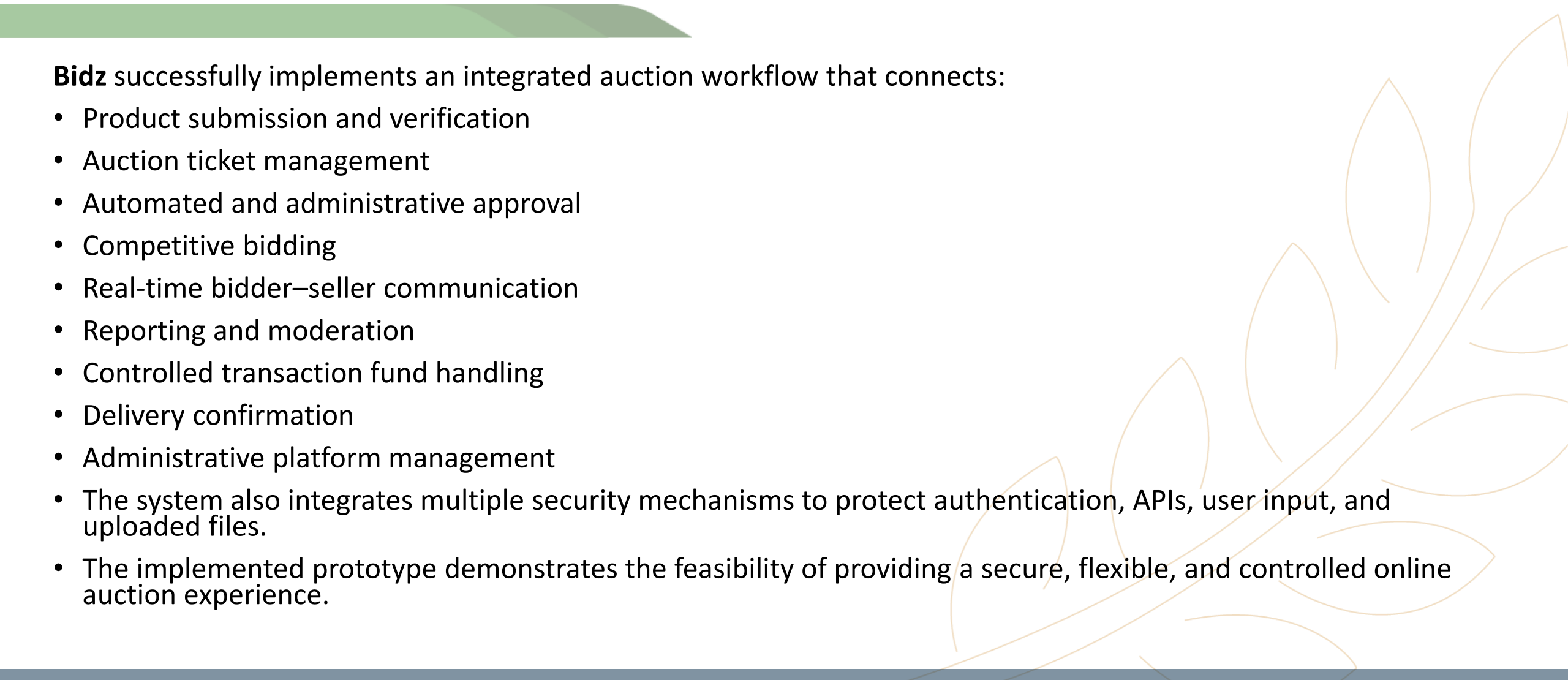


Methodology



- The development of **Bidz** followed a structured software development process:
- Requirements Analysis: Identifying users, roles, business rules, auction lifecycle, and security requirements.
- System Design: Designing the application architecture, database models, authentication flow, and auction workflow.
- Development: Implementing the frontend, backend APIs, auction system, bidding mechanism, real-time chat, wallet flow, and administrative dashboard.
- Security Integration: Applying authentication, authorization, input validation, sanitization, rate limiting, secure cookies, and file upload restrictions.
- Testing: Testing APIs and application functionality using Postman and manual testing.

Results

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Bidz successfully implements an integrated auction workflow that connects:

- Product submission and verification
- Auction ticket management
- Automated and administrative approval
- Competitive bidding
- Real-time bidder–seller communication
- Reporting and moderation
- Controlled transaction fund handling
- Delivery confirmation
- Administrative platform management
- The system also integrates multiple security mechanisms to protect authentication, APIs, user input, and uploaded files.
- The implemented prototype demonstrates the feasibility of providing a secure, flexible, and controlled online auction experience.

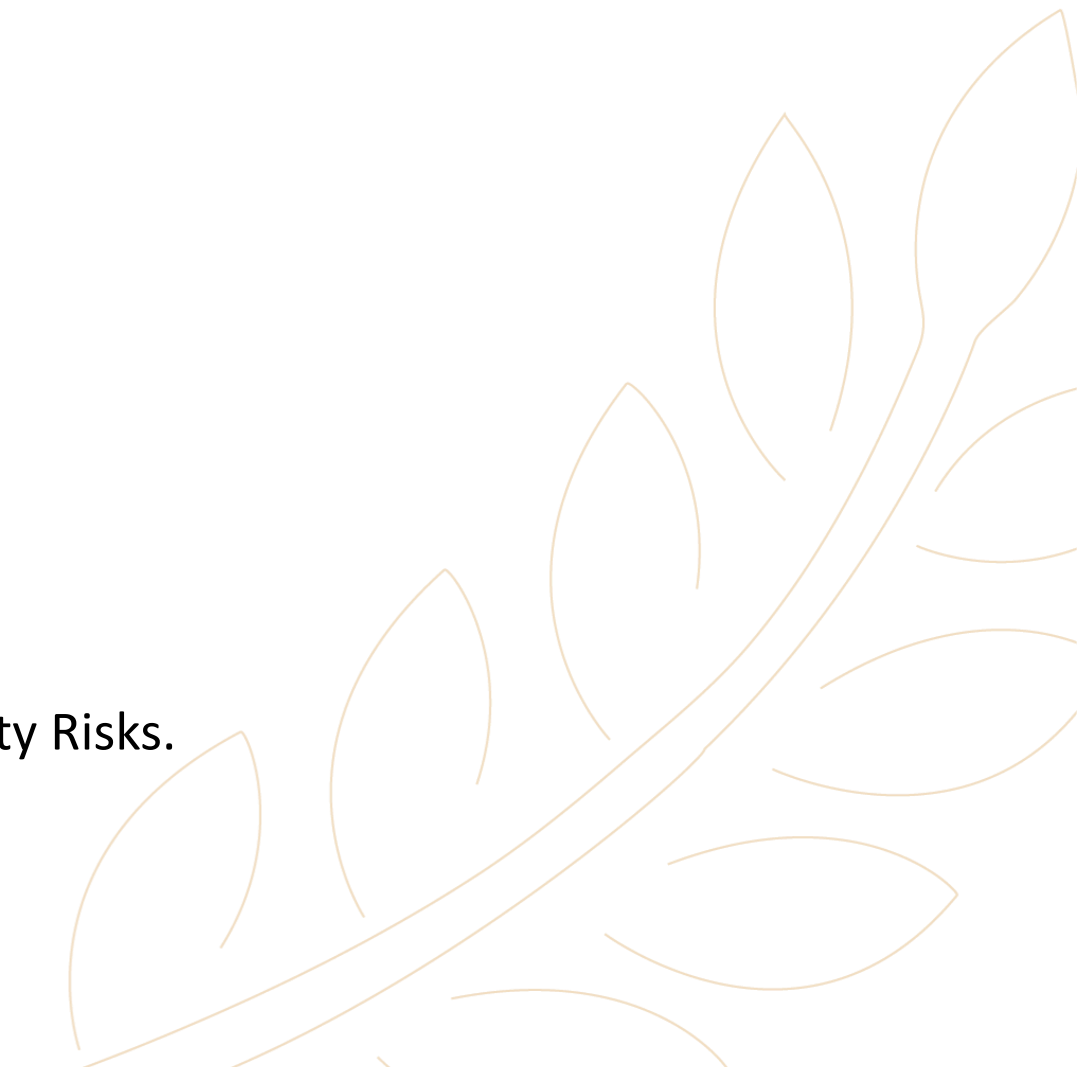
Future Work

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- Future development of **Bidz** could include:
 - Integration with real online payment gateways.
 - Advanced fraud and suspicious activity detection.
 - Enhanced product and seller verification.
 - AI-based product and auction recommendations.
 - Integrated delivery services.
 - Mobile application.
 - Advanced analytics and auction insights.
 - Enhanced dispute resolution.
 - Rating and review system.
 - These improvements could further enhance the platform's security, scalability, trust, and user experience.
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References

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 - [Node.js Documentation](#). [Node.js Foundation](#).
 - [Express.js Documentation](#). [OpenJS Foundation](#).
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 - [Socket.IO Documentation](#).
 - [JWT.io](#). [JSON Web Tokens Introduction](#).
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 - [Helmet.js Documentation](#).
 - [Postman Documentation](#). [Postman, Inc.](#)
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- A decorative graphic in the bottom right corner consisting of several stylized orange leaves and a curved line, resembling a branch or a fan.

The image features a stylized background. On the left, there are overlapping curved shapes in shades of green and grey. On the right, a thin, light brown line depicts a branch with several elongated, pointed leaves. The text "Thank You" is centered in the white space between these elements.

Thank You